

SPARWOOD, BC, Canada

WMO: 717820

Lat: 49.7450N

Lon: 114.8839W

Elev: 3734

StdP: 12.82

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-15.8	-9.6	-22.3	1.8	-14.6	-16.2	2.6	-9.2	21.0	11.6	18.0	13.0	5.1	10	0.642

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.5	85.7	60.4	82.1	59.4	78.6	58.1	62.4	79.7	60.7	77.7	59.2	75.5	8.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.5	78.3	65.9	54.6	72.7	64.2	52.7	68.0	63.1	29.9	79.9	28.6	77.9	27.5	75.8	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	14.3	12.7	DB	-23.9	90.6	5.5	3.5	-27.8	93.2	-31.0	95.3	-34.1	97.2	-38.1	99.8
			WB	-24.3	65.8	5.5	2.9	-28.3	67.9	-31.5	69.5	-34.6	71.2	-38.6	73.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	39.9	20.3	22.8	31.5	39.7	48.1	54.5	61.8	60.2	51.5	39.8	28.5	19.2	(6)
(7)		DBStd	17.22	13.61	11.37	9.43	6.44	6.50	5.57	5.43	5.29	6.52	7.40	10.81	12.55	(7)
(8)		HDD50	4687	920	762	574	313	116	19	2	2	59	323	644	953	(8)
(9)		HDD65	9196	1385	1182	1039	759	525	318	126	163	406	781	1094	1418	(9)
(10)		CDD50	1010	0	0	0	3	57	154	369	317	103	7	0	0	(10)
(11)		CDD65	45	0	0	0	0	0	2	28	14	1	0	0	0	(11)
(12)		CDH74	1874	0	0	0	1	41	152	873	687	120	0	0	0	(12)
(13)		CDH80	520	0	0	0	0	3	30	273	196	18	0	0	0	(13)
(14)	Wind	WSAvg	5.3	5.6	5.7	6.1	5.6	5.1	5.0	5.0	4.8	4.8	5.1	5.5	5.1	(14)
(15)	Precipitation	PrecAvg	33.8	3.3	2.3	2.6	2.1	3.0	3.4	1.9	1.8	2.3	3.5	4.0	3.5	(15)
(16)		PrecMax	46.5	5.7	4.5	6.5	3.7	5.9	7.2	5.6	5.8	6.9	6.2	10.4	6.8	(16)
(17)		PrecMin	25.9	0.5	0.6	1.0	1.1	1.1	2.1	0.0	0.5	0.3	1.7	0.6	1.9	(17)
(18)		PrecStd	5.9	1.2	1.1	1.4	0.7	1.3	1.3	1.4	1.2	1.5	1.5	2.8	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.6	47.3	57.6	70.1	78.6	84.4	90.6	89.5	82.8	67.8	53.8	42.8	(19)
(20)		MCWB	40.3	38.3	42.8	50.2	55.6	60.2	62.7	59.9	58.4	52.4	44.0	39.4	(20)	
(21)		2%	DB	41.1	42.5	50.9	63.8	73.9	78.3	86.7	85.0	77.5	61.6	48.7	39.5	(21)
(22)		MCWB	36.7	36.3	40.5	47.7	53.5	58.1	60.7	59.8	56.6	49.0	43.6	36.1	(22)	
(23)		5%	DB	38.5	39.7	47.1	58.2	69.0	74.2	83.0	81.7	72.7	56.7	45.1	37.4	(23)
(24)		MCWB	34.8	34.9	39.0	44.2	51.8	55.8	60.1	59.0	55.0	46.6	40.5	34.5	(24)	
(25)		10%	DB	36.4	37.1	43.6	53.1	63.9	69.6	79.2	77.6	67.3	52.3	41.9	35.0	(25)
(26)		MCWB	33.4	33.0	37.2	41.9	49.5	54.2	58.9	57.8	52.6	44.2	38.1	32.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.0	40.1	44.7	51.5	57.2	62.1	65.5	64.0	60.2	54.2	47.4	40.1	(27)
(28)		MADB	44.8	44.0	52.6	67.0	74.3	79.8	84.5	81.9	75.0	64.2	50.9	42.0	(28)	
(29)		2%	WB	37.5	37.5	41.9	48.2	54.8	59.5	63.2	61.6	57.5	50.4	43.8	36.8	(29)
(30)		MADB	40.2	41.6	48.7	62.2	70.3	75.4	80.4	79.6	74.1	59.2	48.0	38.6	(30)	
(31)		5%	WB	35.3	35.3	39.9	45.3	52.7	57.2	61.5	60.1	55.6	47.7	40.9	34.7	(31)
(32)		MADB	37.7	39.0	46.2	56.1	66.2	71.0	78.2	77.7	70.6	55.3	44.6	37.0	(32)	
(33)		10%	WB	33.6	33.3	37.7	42.8	50.4	55.1	59.9	58.6	53.5	44.9	38.4	32.7	(33)
(34)		MADB	36.0	36.7	43.0	51.9	62.3	66.6	75.9	75.2	65.9	51.2	41.8	35.0	(34)	
(35)	Mean Daily Temperature Range	MDBR	14.7	17.9	17.5	21.0	23.6	23.7	29.5	30.5	26.4	19.7	13.6	13.1	(35)	
(36)		5% DB	MADB	12.6	17.5	22.8	33.7	36.0	34.9	37.9	39.5	38.9	28.4	15.8	10.2	(36)
(37)		MCWBR	9.5	12.2	13.8	18.5	17.8	16.2	15.4	16.8	19.4	16.8	11.0	8.2	(37)	
(38)		5% WB	MADB	11.5	15.7	20.7	30.6	32.3	30.9	33.5	34.9	36.2	26.3	14.7	10.2	(38)
(39)		MCWBR	9.2	11.7	13.2	17.4	16.4	14.7	14.5	15.9	18.6	16.2	11.0	8.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.246	0.258	0.283	0.328	0.345	0.333	0.336	0.344	0.312	0.292	0.267	0.238	(40)	
(41)		taud	2.361	2.334	2.315	2.271	2.389	2.476	2.482	2.461	2.532	2.565	2.488	2.389	(41)	
(42)		Ebn at Noon	262	286	296	291	290	293	291	283	283	272	253	250	(42)	
(43)		Edh at Noon	20	27	33	38	36	33	33	32	27	22	18	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	349	655	989	1425	1685	1812	2115	1742	1212	728	368	273	(44)	
(45)		RadStd	32	51	87	111	142	148	168	127	129	78	35	23	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+101	N/A
(47) Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page